

# Daily Logic Drill #4 — Answers & Investigations

Sheet by David Akinyele-Aje

| Section | Topic                    | Questions | Target time  |
|---------|--------------------------|-----------|--------------|
| A       | Rapid Recognition        | 10        | 2 min 30 sec |
| B       | Manipulation Drills      | 10        | 8 min        |
| C       | Substitution & Structure | 8         | 10 min       |
| D       | Challenge                | 5         | 15 min       |

New toolkit today: Disproof by Counterexample · Boundary Case Discovery · Derivative & Function Counterexamples · Universal Refutation.

## Section A

1. Provide the minimal positive integer counterexample  $n$  to: “For all positive integers  $n$ , if  $n$  is prime, then  $n^2 + 2$  is not prime.” (after TMUA 2022 Paper 2 Q3)

**Answer:** 3

**Method:** Test small primes: For  $n = 2$ ,  $2^2 + 2 = 6$  (composite, not a counterexample). For  $n = 3$ , 3 is prime and  $3^2 + 2 = 11$ , which is prime! Thus  $n = 3$  satisfies the premise but falsifies the conclusion, making it the minimal counterexample.

**Investigate further:** For any prime  $p > 3$ ,  $p \equiv \pm 1 \pmod{3}$ , so  $p^2 \equiv 1 \pmod{3}$ , which means  $p^2 + 2 \equiv 0 \pmod{3}$ , so  $p^2 + 2$  is always a multiple of 3 and thus composite. Hence  $n = 3$  is the *only* prime counterexample!

2. Find the smallest positive integer  $a$  for which there exist positive integers  $b, c$  such that  $a \mid bc$  but  $a \nmid b$  and  $a \nmid c$ . (after TMUA 2021 Paper 2 Q4)

**Answer:** 4

**Method:** If  $a$  is prime, Euclid’s Lemma guarantees  $a \mid bc \implies a \mid b \vee a \mid c$ . Thus  $a$  must be composite. The smallest composite number is  $a = 4$ . Taking  $b = 2$  and  $c = 2$ , we have  $bc = 4$ , so  $4 \mid 4$ , but  $4 \nmid 2$ . Thus  $a = 4$  is the minimal counterexample.

**Investigate further:** If  $a$  must be squarefree, what is the smallest counterexample?  $a = 6$ , with  $b = 2, c = 3$ .

3. Provide a function  $f(x)$  with  $f'(x) > 0$  for all  $x \in \mathbb{R}$  that serves as a counterexample to: “If  $f'(x) > 0$  for all  $x$ , then  $f(x) > 0$  for all  $x$ .” (after TMUA 2018 Paper 2 Q6)

**Answer:**  $f(x) = x$

**Method:** The linear function  $f(x) = x$  has  $f'(x) = 1 > 0$  for all  $x \in \mathbb{R}$ , but  $f(x) \leq 0$  for all  $x \leq 0$  (e.g.  $f(-5) = -5 \not> 0$ ). Another valid counterexample is  $f(x) = e^x - 2$ .

**Investigate further:** A positive derivative guarantees that the function is strictly increasing, not that its values are positive.

4. Find a real number  $x > 0$  that serves as a counterexample to: “ $x > \frac{1}{x}$  for all  $x > 0$ .”

**Answer:**  $\frac{1}{2}$

**Method:** For  $x = \frac{1}{2} > 0$ , we have  $\frac{1}{x} = 2$ . Since  $\frac{1}{2} < 2$ , the inequality fails. Any  $x \in (0, 1]$  disproves the claim.

**Investigate further:** The inequality  $x > 1/x$  on positive reals holds if and only if  $x > 1$ .

5. Provide a pair of real numbers  $(x, y)$  that serves as a counterexample to: “For all  $x, y \in \mathbb{R}$ , if  $x^2 > y^2$ , then  $x > y$ .”

**Answer:**  $(-3, 1)$

**Method:** Let  $x = -3$  and  $y = 1$ . Then  $x^2 = 9$  and  $y^2 = 1$ , so  $x^2 > y^2$  holds. However,  $-3 < 1$ , so  $x > y$  is false.

**Investigate further:** Squaring preserves inequality only when both numbers are non-negative.

6. Find the smallest positive integer  $n$  for which  $2^n - 1$  is composite despite  $n$  being prime.

**Answer:** 11

**Method:** Primes  $n$ :  $2^2 - 1 = 3$  (prime),  $2^3 - 1 = 7$  (prime),  $2^5 - 1 = 31$  (prime),  $2^7 - 1 = 127$  (prime). For  $n = 11$ :  $2^{11} - 1 = 2047 = 23 \times 89$ , which is composite. Thus  $n = 11$  is the smallest counterexample.

**Investigate further:** Mersenne numbers  $2^p - 1$  need not be prime even when  $p$  is prime.

7. Provide a pair of non-negative real numbers  $(a, b)$  that serves as a counterexample to: “ $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$ .”

**Answer:**  $(1, 1)$

**Method:** For  $a = 1, b = 1$ ,  $\sqrt{a+b} = \sqrt{2}$ , whereas  $\sqrt{a} + \sqrt{b} = 1 + 1 = 2$ . Since  $\sqrt{2} \neq 2$ , the identity fails.

**Investigate further:** The equality  $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$  holds if and only if  $ab = 0$ .

8. Find the smallest positive integer  $n$  that serves as a counterexample to: “ $n^2 + 1$  is always prime.”

**Answer:** 3

**Method:** For  $n = 1$ ,  $1^2 + 1 = 2$  (prime). For  $n = 2$ ,  $2^2 + 1 = 5$  (prime). For  $n = 3$ ,  $3^2 + 1 = 10 = 2 \times 5$ , which is composite. Smallest  $n$  is 3.

**Investigate further:** For which other small integers is  $n^2 + 1$  composite?  $n = 5 \implies 26 = 2 \times 13$ ,  $n = 7 \implies 50$ .

9. Provide a differentiable function  $f : [1, \infty) \rightarrow \mathbb{R}$  with  $\lim_{x \rightarrow \infty} f'(x) = 0$  that disproves: “If  $\lim_{x \rightarrow \infty} f'(x) = 0$ , then  $\lim_{x \rightarrow \infty} f(x)$  exists and is finite.”

**Answer:**  $f(x) = \sqrt{x}$

**Method:** For  $f(x) = \sqrt{x}$ ,  $f'(x) = \frac{1}{2\sqrt{x}} \rightarrow 0$  as  $x \rightarrow \infty$ , but  $f(x) = \sqrt{x} \rightarrow \infty$ . Another standard counterexample is  $f(x) = \ln x$ , where  $f'(x) = 1/x \rightarrow 0$  while  $\ln x \rightarrow \infty$ .

**Investigate further:** A vanishing derivative means the growth rate slows to zero, but the total accumulated growth can still diverge to infinity.

10. Find the smallest odd composite integer  $n$  that is not a prime power.

**Answer:** 15

**Method:** Odd composites in order:  $9 = 3^2$  (prime power),  $15 = 3 \times 5$  (not a prime power). Thus 15 is the smallest.

**Investigate further:** Any non-prime-power odd composite has at least two distinct odd prime factors.

## Section B

1. A student claims: “If a polynomial  $f(x)$  satisfies  $f(n) \in \mathbb{Z}$  for all  $n \in \mathbb{Z}$ , then  $f'(n) \in \mathbb{Z}$  for all  $n \in \mathbb{Z}$ .” Provide a quadratic polynomial  $f(x)$  that serves as a counterexample. (after TMUA 2017 Paper 2 Q16)

**Answer:**  $f(x) = \frac{x(x-1)}{2}$

**Method:** For  $f(x) = \frac{x^2-x}{2} = \binom{x}{2}$ , the product of two consecutive integers  $n(n-1)$  is always even, so  $f(n) \in \mathbb{Z}$  for all  $n \in \mathbb{Z}$ . However,  $f'(x) = x - \frac{1}{2}$ . For any integer  $n$ ,  $f'(n) = n - \frac{1}{2} \notin \mathbb{Z}$ . This disproves the claim.

**Investigate further:** This counterexample relies on the binomial basis polynomials  $\binom{x}{k}$ , which are integer-valued on  $\mathbb{Z}$  but do not have integer coefficients.

2. Give a pair of irrational numbers  $a, b$  whose sum is rational.

**Answer:**  $a = \sqrt{2}, b = -\sqrt{2}$

**Method:**  $\sqrt{2}$  and  $-\sqrt{2}$  are both irrational, but their sum is  $\sqrt{2} + (-\sqrt{2}) = 0 \in \mathbb{Q}$ .

**Investigate further:** Another pair:  $a = 1 + \sqrt{3}, b = 1 - \sqrt{3}$ , where  $a + b = 2$ .

3. Give a pair of irrational numbers  $a, b$  whose product is rational.

**Answer:**  $a = \sqrt{2}, b = \sqrt{2}$

**Method:**  $\sqrt{2}$  is irrational, but  $ab = \sqrt{2} \times \sqrt{2} = 2 \in \mathbb{Q}$ .

**Investigate further:** A pair of distinct positive irrationals:  $a = \sqrt{8}, b = \sqrt{2}$ , where  $ab = \sqrt{16} = 4$ .

4. Provide a pair of functions  $f, g : [0, \infty) \rightarrow \mathbb{R}$  such that  $f(x) \geq g(x)$  for all  $x \geq 0$ , but  $f'(0) < g'(0)$ . (after TMUA 2021 Paper 2 Q12)

**Answer:**  $f(x) = 10, g(x) = x$

**Method:** For  $x \in [0, 5]$ ,  $f(x) = 10 \geq x = g(x)$ . To make  $f(x) \geq g(x)$  for all  $x \geq 0$ , let  $f(x) = 10$  and  $g(x) = 10 - e^{-x}$ . Then  $f(x) = 10 > g(x)$  for all  $x \geq 0$ , but  $f'(0) = 0$  while  $g'(0) = 1 > 0$ , so  $f'(0) < g'(0)$ .

**Investigate further:** Inequalities between functions cannot be differentiated; a function can sit entirely above another while having a smaller derivative.

5. Find the smallest positive integer  $n$  that is a counterexample to: “ $n^2 + n + 41$  is prime for all positive integers  $n$ .”

**Answer:** 40

**Method:** For  $n = 40$ ,  $n^2 + n + 41 = 40^2 + 40 + 41 = 40(41) + 41 = 41(40 + 1) = 41^2 = 1681$ , which is composite ( $41 \times 41$ ). For all  $1 \leq n \leq 39$ ,  $n^2 + n + 41$  is prime. Thus  $n = 40$  is the minimal counterexample.

**Investigate further:** At  $n = 41$ ,  $41^2 + 41 + 41 = 41 \times 43$ , which is also composite, but 40 is smaller.

6. Provide a counterexample of three finite sets  $A, B, C$  showing that  $A \cup B = A \cup C$  does not imply  $B = C$ .

**Answer:**  $A = \{1\}, B = \{1\}, C = \emptyset$

**Method:** Here  $A \cup B = \{1\} \cup \{1\} = \{1\}$ , and  $A \cup C = \{1\} \cup \emptyset = \{1\}$ . The unions are equal, but  $B = \{1\} \neq \emptyset = C$ .

**Investigate further:** Union is not an invertible operation: elements absorbed by  $A$  lose their distinguishing presence in  $B$  or  $C$ .

7. Find the smallest positive integer  $n$  for which  $3^n + 2$  is composite.

**Answer:** 5

**Method:** Test values of  $n$ :  $n = 1 \implies 3^1 + 2 = 5$  (prime);  $n = 2 \implies 3^2 + 2 = 11$  (prime);  $n = 3 \implies 3^3 + 2 = 29$  (prime);  $n = 4 \implies 3^4 + 2 = 83$  (prime);  $n = 5 \implies 3^5 + 2 = 243 + 2 = 245 = 5 \times 49$  (composite). The smallest positive integer is  $n = 5$ .

**Investigate further:** Modulo 5:  $3^1 \equiv 3, 3^2 \equiv 4, 3^3 \equiv 2, 3^4 \equiv 1$ , so  $3^5 \equiv 3 \implies 3^5 + 2 \equiv 5 \equiv 0 \pmod{5}$ .

8. A student conjectures: “For every odd prime  $p$ , the integer  $2^p - p$  is divisible by either 3 or 5.” Find the smallest odd prime  $p$  that serves as a counterexample. (after TMUA 2021 Paper 2 Q10)

**Answer:** 7

**Method:** Test odd primes:

- $p = 3$ :  $2^3 - 3 = 8 - 3 = 5$  (divisible by 5).
- $p = 5$ :  $2^5 - 5 = 32 - 5 = 27$  (divisible by 3).
- $p = 7$ :  $2^7 - 7 = 128 - 7 = 121 = 11^2$ . 121 is divisible by neither 3 nor 5.

Thus  $p = 7$  is the smallest odd prime counterexample.

**Investigate further:** Small patterns in modular arithmetic often break precisely at  $p = 7$ , the first prime where neither  $2^p \equiv p \pmod{3}$  nor  $2^p \equiv p \pmod{5}$  holds.

9. Provide integers  $a, b$  and a modulus  $m > 1$  showing that  $a^2 \equiv b^2 \pmod{m}$  does not imply  $a \equiv \pm b \pmod{m}$ .

**Answer:**  $m = 8, a = 1, b = 3$

**Method:** For  $m = 8, a = 1, b = 3$ :  $a^2 = 1 \equiv 1 \pmod{8}$  and  $b^2 = 9 \equiv 1 \pmod{8}$ , so  $a^2 \equiv b^2 \pmod{8}$ . However,  $3 \not\equiv 1 \pmod{8}$  and  $3 \not\equiv -1 \equiv 7 \pmod{8}$ .

**Investigate further:** The failure occurs because 8 is composite; in  $\mathbb{Z}/8\mathbb{Z}$ , the polynomial  $x^2 - 1$  has four roots  $\{1, 3, 5, 7\}$ .

10. Give a counterexample function  $f : [0, 1] \rightarrow [0, 1]$  showing that a continuous function need not have a UNIQUE fixed point.

**Answer:**  $f(x) = x$

**Method:** The identity function  $f(x) = x$  maps  $[0, 1]$  continuously to  $[0, 1]$ . Every single  $x \in [0, 1]$  is a fixed point, meaning there are infinitely many fixed points rather than a unique one.

**Investigate further:** Brouwer's Fixed Point Theorem guarantees the *existence* of at least one fixed point, but never uniqueness.

## Section C

1. Consider the statement: “(\*) If a whole number  $n$  is 2 more or 4 more than a multiple of 5, then  $n$  is prime.” How many counterexamples to (\*) are there in the range  $1 \leq n \leq 50$ ? (after TMUA 2019 Paper 2 Q4)

**Answer:** 13

**Method:** Numbers of the form  $5k + 2$  in  $[1, 50]$  are: 2, 7, 12, 17, 22, 27, 32, 37, 42, 47. The composites among these are 12, 22, 27, 32, 42 (5 counterexamples).

Numbers of the form  $5k + 4$  in  $[1, 50]$  are: 4, 9, 14, 19, 24, 29, 34, 39, 44, 49. The composites among these are 4, 9, 14, 24, 34, 39, 44, 49 (8 counterexamples).

Total counterexamples =  $5 + 8 = 13$ .

**Investigate further:** Note that 2 and 7 satisfy the premise and are prime, so they are not counterexamples.

2. Which of the following values of  $n$  is a counterexample to the claim: “For all positive integers  $n$ , if  $n$  is prime, then  $n^3 - n$  is NOT divisible by 24?” (after TMUA 2022 Paper 2 Q3)

**Answer:** E) Both  $n = 5$  and  $n = 7$ .

**Method:** Factor  $n^3 - n = (n - 1)n(n + 1)$ . For any prime  $p \geq 5$ ,  $p$  is odd, so  $p - 1$  and  $p + 1$  are consecutive even integers, one of which is divisible by 4, so their product is divisible by 8. Furthermore, among  $p - 1, p, p + 1$ , exactly one is divisible by 3; since  $p \geq 5$  is prime,  $3 \nmid p$ , so  $3 \mid (p - 1)(p + 1)$ . Thus for any prime  $p \geq 5$ ,  $24 \mid (p^3 - p)$ . Both  $n = 5$  ( $5^3 - 5 = 120 = 24 \times 5$ ) and  $n = 7$  ( $7^3 - 7 = 336 = 24 \times 14$ ) are divisible by 24, refuting the claim. Option E.

**Investigate further:** This means every single prime  $p \geq 5$  is a counterexample to the claim!

3. A student claims: “If  $f : \mathbb{R} \rightarrow \mathbb{R}$  is twice differentiable,  $f(0) = 0$ , and  $f''(x) > 0$  for all  $x \in \mathbb{R}$ , then  $f(x) \geq 0$  for all  $x \geq 0$ .” Which of the following functions serves as a counterexample? (after TMUA 2018 Paper 2 Q6)

**Answer:** C)  $f(x) = x^2 - 3x$ .

**Method:** Check the conditions for  $f(x) = x^2 - 3x$ :  $f(0) = 0$ ;  $f'(x) = 2x - 3$ ;  $f''(x) = 2 > 0$  for all  $x$ . All hypotheses hold. However, for  $x \in (0, 3)$ ,  $f(x) = x(x - 3) < 0$ . For instance,  $f(1) = -2 < 0$ . This disproves the claim. Option C.

**Investigate further:** A positive second derivative means the graph is convex (bending upward), but if the initial slope  $f'(0)$  is negative, the function dips below zero before rising.

4. Consider the statement: “For all positive integers  $a, b, c$ , if  $a \mid bc$ , then  $a \mid b$  or  $a \mid c$ .” Which of the following triples  $(a, b, c)$  is a valid counterexample? (after TMUA 2021 Paper 2 Q4)

**Answer:** B) (6, 4, 9).

**Method:** Check option B:  $bc = 4 \times 9 = 36$ .  $a = 6$  divides 36 ( $36/6 = 6$ ). However,  $6 \nmid 4$  and  $6 \nmid 9$ . Both disjuncts fail while the premise holds. Thus (6, 4, 9) is a valid counterexample. In option A,  $a = 2 \mid 4$ ; in C,  $a = 5 \mid 10$ ; in D,  $a = 7 \mid 14$ .

**Investigate further:** A counterexample requires  $a$  to share distinct prime factors between  $b$  and  $c$ .

5. Which of the following polynomials satisfies  $f(n) \in \mathbb{Z}$  for all  $n \in \mathbb{Z}$ , but  $f'(0) \notin \mathbb{Z}$ ? (after TMUA 2017 Paper 2 Q16)

**Answer:** D) Both B and C.

**Method:** For B:  $f(x) = \frac{x^2+x}{2} = \binom{x+1}{2}$ , integer-valued on  $\mathbb{Z}$ .  $f'(x) = x + \frac{1}{2} \implies f'(0) = \frac{1}{2} \notin \mathbb{Z}$ .  
For C:  $f(x) = \frac{x^3-x}{6} = \frac{(x-1)x(x+1)}{6} = \binom{x+1}{3}$ , integer-valued on  $\mathbb{Z}$  since the product of three consecutive integers is always divisible by 6.  $f'(x) = \frac{3x^2-1}{6} \implies f'(0) = -\frac{1}{6} \notin \mathbb{Z}$ .  
Both B and C satisfy the condition. Option D.

**Investigate further:** Every integer-valued polynomial is an integer linear combination of the binomial polynomials  $\binom{x}{k}$ .

6. A student conjectures: “If a continuous function  $f : [-\pi, \pi] \rightarrow \mathbb{R}$  satisfies  $\int_{-\pi}^{\pi} f(x)dx = 0$ , then  $f(x) = 0$  for all  $x \in [-\pi, \pi]$ .” Which of the following is a valid counterexample? (after TMUA 2019 Paper 2 Q14)

**Answer:** D) All of A, B, C.

**Method:** Evaluate the integral for each function on  $[-\pi, \pi]$ :

- A:  $\int_{-\pi}^{\pi} \cos x \, dx = [\sin x]_{-\pi}^{\pi} = 0 - 0 = 0$ , but  $\cos 0 = 1 \neq 0$ .
- B:  $\int_{-\pi}^{\pi} \sin x \, dx = [-\cos x]_{-\pi}^{\pi} = 1 - 1 = 0$ , but  $\sin(\pi/2) = 1 \neq 0$ .
- C:  $\int_{-\pi}^{\pi} x \, dx = \left[\frac{x^2}{2}\right]_{-\pi}^{\pi} = 0$ , but  $f(1) = 1 \neq 0$ .

All three functions have zero integral yet are not identically zero. Option D.

**Investigate further:** Any non-trivial odd function on a symmetric interval  $[-a, a]$  integrates to zero.

7. Consider two statements about functions  $f, g : [0, \infty) \rightarrow \mathbb{R}$  with  $f(0) = g(0) = 0$ :

- $f'(x) \geq g'(x)$  for all  $x \geq 0 \implies f(x) \geq g(x)$  for all  $x \geq 0$ .
- $f(x) \geq g(x)$  for all  $x \geq 0 \implies f'(x) \geq g'(x)$  for all  $x \geq 0$ .

Which of I and II are true? (after TMUA 2021 Paper 2 Q12)

**Answer:** B) I only.

**Method:** By the Fundamental Theorem of Calculus,  $f(x) - g(x) = \int_0^x (f'(t) - g'(t)) \, dt$ . If  $f'(t) \geq g'(t)$ , the integrand is non-negative, so the integral is  $\geq 0$ , proving  $f(x) \geq g(x)$ . Thus statement I is TRUE.  
For statement II: consider  $f(x) = 2x$  and  $g(x) = x + \sin x$ . For  $x \geq 0$ ,  $f(x) - g(x) = x - \sin x \geq 0$ , so  $f(x) \geq g(x)$ . However,  $f'(x) = 2$  and  $g'(x) = 1 + \cos x$ . At  $x = 0$ ,  $f'(0) = 2 \geq 2$ , but consider  $f(x) = 10x$  and  $g(x) = 10x - 5 \cos(10x) + 5$ : oscillating derivatives frequently exceed  $f'(x)$ . Statement II is FALSE.  
Option B.

**Investigate further:** Integration preserves inequalities; differentiation does not.

8. Consider the theorem: “An odd prime  $p$  can be written as the sum of two integer squares if and only if  $p \equiv 1 \pmod{4}$ .” Which of the following would constitute a counterexample to the **converse** of this theorem? (after TMUA 2018 Paper 2 Q17)

**Answer:** A) A prime  $p \equiv 1 \pmod{4}$  that cannot be written as the sum of two squares.

**Method:** The theorem states  $P \iff Q$ , where  $P$ : “ $p$  is a sum of two squares” and  $Q$ : “ $p \equiv 1 \pmod{4}$ ” (for odd primes). The converse is  $Q \Rightarrow P$ : “If an odd prime  $p \equiv 1 \pmod{4}$ , then  $p$  can be written as the sum of two squares.” A counterexample to  $Q \Rightarrow P$  must satisfy the hypothesis  $Q$  ( $p \equiv 1 \pmod{4}$ ) but fail the conclusion  $P$  (cannot be written as the sum of two squares). Option A.

**Investigate further:** Refuting a conditional statement requires satisfying the antecedent while falsifying the consequent.

## Section D

1. Consider: I. “ $n = 41$  is a counterexample to ‘ $n^2 + n + 41$  is always prime’.” II. “ $n = 40$  is the smallest counterexample to ‘ $n^2 + n + 41$  is always prime’.” III. “ $n = 4$  is a counterexample to ‘ $n^2 + n + 41$  is always prime’.” Which of I, II, III are true?

**Answer:** E) I and II only

**Method:** I:  $41^2 + 41 + 41 = 41(41 + 1 + 1) = 41 \times 43$ , composite. So I is true.

II: For  $1 \leq n \leq 39$ ,  $n^2 + n + 41$  is prime. At  $n = 40$ ,  $40^2 + 40 + 41 = 41^2$ , composite. Thus  $n = 40$  is indeed the smallest positive integer counterexample. II is true.

III: For  $n = 4$ ,  $4^2 + 4 + 41 = 61$ , which is prime. So  $n = 4$  is not a counterexample. III is false.

Hence, I and II only are true.

**Investigate further:** Euler’s polynomial  $n^2 + n + 41$  is the classical demonstration that finite verification cannot prove universal claims.

2. Prove by contradiction: “There do not exist positive integers  $a, b$  with  $a^2 - b^2 = 1$ .”

**Answer:** Proof by contradiction

**Method:** Assume toward a contradiction that there exist positive integers  $a, b$  such that  $a^2 - b^2 = 1$ . Factorising as a difference of two squares gives  $(a - b)(a + b) = 1$ . Since  $a, b \in \mathbb{Z}^+$ , we have  $a \geq 1$  and  $b \geq 1$ , so  $a + b \geq 2$ . However, the only factorisation of 1 into positive integers is  $1 \times 1$ , requiring  $a + b = 1$ . This contradicts  $a + b \geq 2$ . Thus, no such positive integers exist.

**Investigate further:** Do non-zero integer solutions exist if  $a, b$  are not restricted to positive integers?

3. A student claims: “For every positive integer  $n$ , at least one of  $n, n + 2, n + 4$  is divisible by 3.” Determine, with proof, whether this claim is true.

**Answer:** True

**Method:** Proof: Any positive integer  $n$  can be written as  $3k, 3k + 1$ , or  $3k + 2$  for some integer  $k \geq 0$ . Case 1: If  $n = 3k$ , then  $n$  is divisible by 3.

Case 2: If  $n = 3k + 1$ , then  $n + 2 = (3k + 1) + 2 = 3k + 3 = 3(k + 1)$ , which is divisible by 3.

Case 3: If  $n = 3k + 2$ , then  $n + 4 = (3k + 2) + 4 = 3k + 6 = 3(k + 2)$ , which is divisible by 3.

In all cases, at least one of  $n, n + 2, n + 4$  is divisible by 3. The claim is true.

**Investigate further:** Notice that  $n, n+2, n+4 \equiv n, n-1, n+1 \pmod{3}$ , so they cover all three residue classes modulo 3.

4. Give a counterexample to: “If  $p$  is prime, then  $2^p + 1$  is prime.”

**Answer:** 3

**Method:** For  $p = 3$  (which is prime),  $2^p + 1 = 2^3 + 1 = 8 + 1 = 9 = 3^2$ , which is composite. (Also for  $p = 5$ ,  $2^5 + 1 = 33 = 3 \times 11$ ).

**Investigate further:** For any odd prime  $p$ ,  $2^p + 1 = (2 + 1)(2^{p-1} - 2^{p-2} + \dots + 1) = 3 \times M$ , so  $2^p + 1$  is always a multiple of 3, and hence composite for all  $p > 1$  with  $2^p + 1 > 3$ !

5. Statement (\*): “For every positive integer  $n \geq 2$ , there is a prime strictly between  $n$  and  $2n$ .”  
 (a) Verify (\*) for  $n = 2, 3, 4, 5$  by direct computation. (b) Does this verification constitute a proof of (\*)? Justify your answer.

**Answer:** (a)  $n = 2 : 3$ ;  $n = 3 : 5$ ;  $n = 4 : 5$  or  $7$ ;  $n = 5 : 7$ . (b) No, finite checking does not prove a universal statement.

**Method:** (a) For  $n = 2$ , prime  $3 \in (2, 4)$ . For  $n = 3$ , prime  $5 \in (3, 6)$ . For  $n = 4$ , prime  $5 \in (4, 8)$ . For  $n = 5$ , prime  $7 \in (5, 10)$ .

(b) No. Verification for finitely many cases ( $n = 2, 3, 4, 5$ ) does not constitute a proof for infinitely many positive integers  $n \geq 2$ . A single unchecked value could be a counterexample. (The statement is true for all  $n \geq 2$ , known as Bertrand’s Postulate, but proving it requires a general argument).

**Investigate further:** Bertrand’s Postulate was first proved by Chebyshev in 1852 using properties of the gamma function and prime factorisations of central binomial coefficients.

## Top 5 Patterns Today

- Refuting Implications by Finding Witness Instances.** To disprove  $\forall x, (P(x) \Rightarrow Q(x))$ , find a single counterexample where  $P(x)$  holds while  $Q(x)$  fails. (see A1, A2, A3, A4, B8)
- Derivative and Function Inversion Traps.** A function satisfying  $f'(x) > 0$  everywhere does not need to be positive ( $f(x) = x$ ); similarly,  $f(x) \geq g(x)$  does not force  $f'(x) \geq g'(x)$ . (see A3, A9, B4, C3, C7)
- Integer-Valued Polynomial Derivatives.** A polynomial integer-valued on  $\mathbb{Z}$  (such as binomial coefficients  $\binom{x}{2} = \frac{x^2-x}{2}$ ) need not have integer coefficients or integer derivatives. (see B1, C5)
- Prime Generator Boundaries and Divisibility.** Conjectures asserting primality (like  $n^2 + n + 41$  or  $2^p - p$ ) fail at structural boundaries ( $n = 40, p = 7$ ) where algebraic factoring or non-residues intervene. (see A6, A8, B5, B7, D1)
- Non-Trivial Modular Residue Square Roots.** In non-prime moduli (like  $m = 8$ ), the congruence  $x^2 \equiv 1 \pmod{8}$  has four distinct roots ( $\pm 1, \pm 3$ ), breaking the field property  $a^2 \equiv b^2 \implies a \equiv \pm b$ . (see A10, B9, C4)

## Common Traps to Avoid

- Testing Only Positive or Symmetrical Values.** Order relations like  $x^2 > y^2 \implies x > y$  fail only when considering negative values ( $x = -3, y = 1$ ); always check signed edge cases. (see



$A4, A5, A7, B2$ )

2. **Confusing Implication Refutation with Converse Refutation.** Refuting the converse of  $P \iff Q$  requires finding a witness satisfying  $Q$  but not  $P$ , rather than attempting to falsify the forward direction. (see C8, A1, B8)
3. **Assuming Derivatives Inherit Function Inequalities.** The height of a curve  $f(x) \geq g(x)$  provides zero information about its slope  $f'(x) \geq g'(x)$ ; slopes can cross while functions remain separated. (see B4, C3, C7)
4. **Assuming Finite Verification Equals Proof.** Checking that a statement holds for hundreds of small cases (Bertrand's Postulate, Euler's polynomial) demonstrates only the absence of small counterexamples, never a proof. (see D5, B5, D1)
5. **Overlooking Composite Prime Factors  $p \leq \sqrt{n}$ .** Refuting that  $a \mid bc \implies a \mid b \vee a \mid c$  requires a composite modulus  $a$  that splits its prime factors across  $b$  and  $c$ . (see A2, C4, B9)

---

*"Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised."*

---

Found a mistake or want to contribute a sheet?

Visit the open-source project at [github.com/The-CerealDev/speed-maths](https://github.com/The-CerealDev/speed-maths)